

DRAFT California State Water Resources Control Board’s Guide to Publishing on the California Open Data Portal

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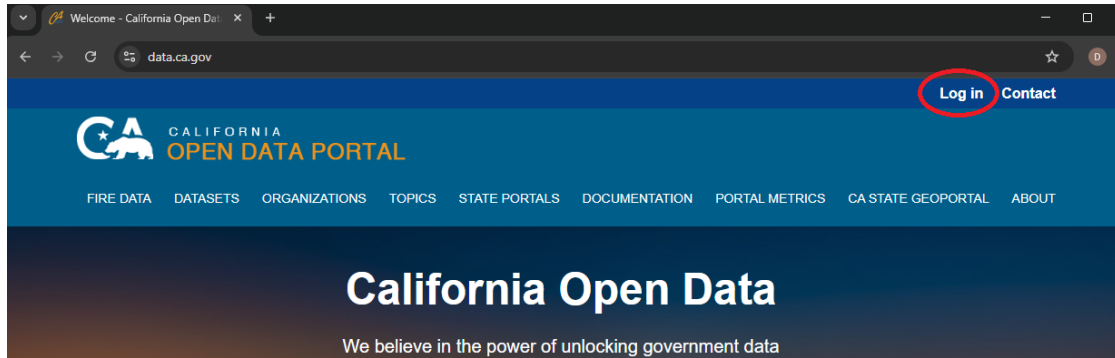
1 Introduction

This document describes the basics of how to publish and maintain a dataset on the [California Open Data Portal](#) for staff at the California State Water Resources Control Board. In the context of the California Open Data Portal, a dataset is a collection of data resources (which are files or links to files containing data, such as CSV files), together with a description and other supporting information (metadata), at a fixed URL.

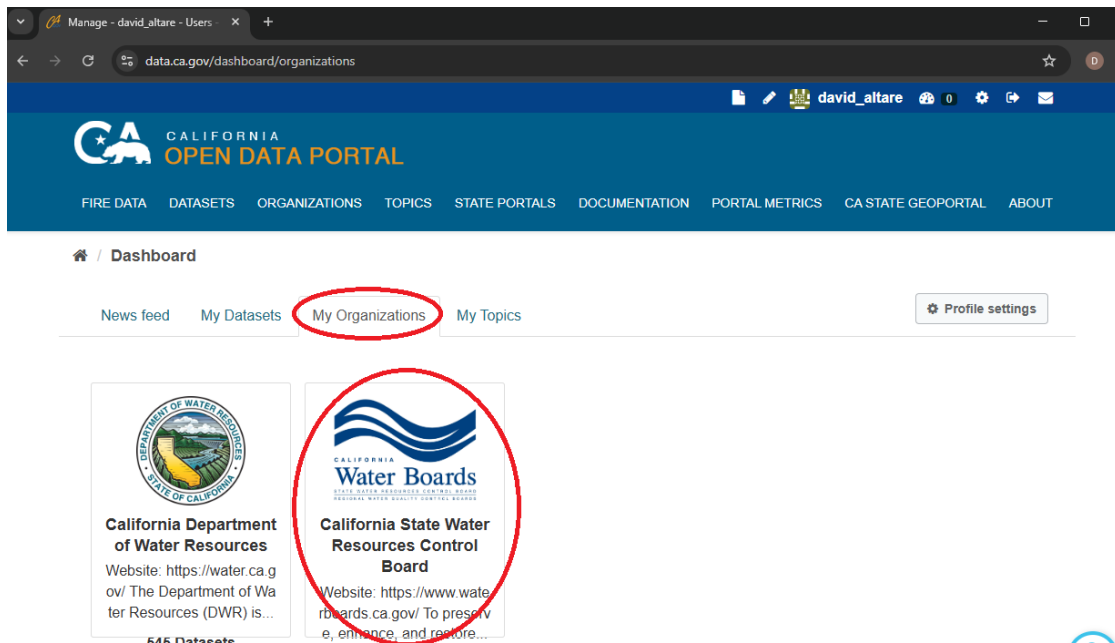
The instructions below demonstrate how to create a new dataset, then add one or more data resources to that dataset. If you want to manage an existing dataset (including adding a new data resource, updating an existing data resource, or editing dataset metadata), skip to the [Managing an Existing Dataset or Data Resource](#) section.

2 Creating a New Dataset

1. Go to <https://data.ca.gov> and click on the **Log in** button at the top right of the window to log in to your user account.



2. In your account dashboard, click on the **My Organizations** tab, then click on the **California State Water Resources Control Board** icon (if you're not already a member of this organization, you'll need to work with Office of Information Management and Analysis (OIMA) staff to be added to it).



3. On the organization page, click on the **Add Dataset** button.

The screenshot shows the 'California State Water Resources Control Board' organization page on the 'data.ca.gov' portal. The page header includes the 'CALIFORNIA OPEN DATA PORTAL' logo and navigation links like 'FIRE DATA', 'DATASETS', 'ORGANIZATIONS', etc. The main content area features the organization's logo, a brief description, and a list of tabs: 'Datasets', 'Activity Stream', and 'About'. The 'Add Dataset' button is highlighted with a red circle. Below the tabs is a search bar and a dropdown menu showing '66 datasets found'. A snippet of a dataset titled 'Drinking Water - SAFER Dashboard Failing and At-Risk Drinking Water Systems' is visible.

4. Fill in the dataset metadata (e.g., Title, Description, etc.). Guidance on how to populate the metadata is available [here](#) and [here](#).

The screenshot shows the 'Create Dataset' form on the 'data.ca.gov' portal. The form is titled 'Create Dataset' and has two steps: '1 Create dataset' and '2 Add data'. The '1 Create dataset' step is active. It includes fields for 'Title' (with a placeholder 'eg. A descriptive title'), 'URL' (with a placeholder 'data.ca.gov/dataset/<dataset>' and an 'Edit' button), and 'Description' (with a placeholder 'eg. Some useful notes about the data'). A note at the bottom states 'You can use Markdown formatting here'.

When you're finished, click on the **Next: Add Data** button at the bottom of the page.

The screenshot shows the 'Create Dataset' form on the California Open Data Portal. The browser address bar displays 'data.ca.gov/dataset/new?group=a652035f-505c-4c24-8464-fed3623acfd'. The form contains three 'Custom Field' sections, each with 'Key' and 'Value' input boxes. Below these fields, a small text block states: 'The data license you select above only applies to the contents of any resource files that you add to this dataset. By submitting this form, you agree to release the metadata values that you enter into the form under the [Open Database License](#).' To the right of this text is a '* Required field' label. At the bottom right of the form, a blue button labeled 'Next: Add Data' is circled in red.

5. Add a file containing your data using either the **Upload** or **Link** buttons, add a title and description for the resource, then click the **Finish** button, or click the **Save & add another** button to add another data resource. Guidance on how to format a data file for publication on the open data portal is available [here](#) and [here](#).

The screenshot shows the 'Add data to the dataset' form on the California Open Data Portal. The browser address bar displays 'data.ca.gov/dataset/example-dataset/resource/new'. The page header includes the 'CALIFORNIA OPEN DATA PORTAL' logo and a navigation menu with links like 'FIRE DATA', 'DATASETS', 'ORGANIZATIONS', 'TOPICS', 'STATE PORTALS', 'DOCUMENTATION', 'PORTAL METRICS', 'CA STATE GEOPORTAL', and 'ABOUT'. Below the header, a breadcrumb trail reads: 'Organizations / California State Water... / Example Dataset / Edit / Add New Resource'. The main content area has a blue header with two steps: '1 Create dataset' and '2 Add data'. On the left, a 'What's a resource?' sidebar explains that a resource can be any file or link to a file containing useful data. The 'File:' section has 'Upload' and 'Link' buttons circled in red. Below this is a 'Title:' input field with the placeholder text 'eg. January 2015 Gold Prices'. The 'Description:' section has a large text area with the placeholder text 'Some useful notes about the data' and a note 'You can use Markdown formatting here'. The 'Format:' section has a dropdown menu with the placeholder text 'eg. CSV, XML or JSON'. At the bottom right, three buttons are visible: 'Previous', 'Save & add another' (circled in red), and 'Finish' (circled in blue).

6. Add a data dictionary for the data resource you just created using one or both of the following methods:

- A: Upload a PDF containing the data dictionary as a separate resource in the dataset
- B: Add the data dictionary directly to the data resource's page (the data dictionary will appear below the data table)

Guidance on how to populate the data dictionary is available [here](#), [here](#), and [here](#).

- A. Upload a PDF containing the data dictionary to the dataset (as a separate resource), by going to the dataset homepage and clicking on the **Manage** button.

The screenshot shows the 'Example Dataset' page on the California Open Data Portal. The page is for a dataset named 'Example Dataset' under the organization 'California State Water Boards'. The 'Manage' button is circled in red. The page includes a 'Data and Resources' section with an 'Example Resource' and a search bar containing the word 'water'. Below this is an 'Additional Info' section with a 'Download' button and a table of metadata.

Example Dataset

Followers: 0

[Follow](#)

Organization

California State Water Resources Control Board

Website: <https://www.waterboards.ca.gov/> To preserve, enhance, and restore the quality of California's water resources and drinking water for the protection

[Dataset](#) [Topics](#) [Showcases](#) [Activity Stream](#) [Manage](#)

Example Dataset PRIVATE

Test description

Data and Resources

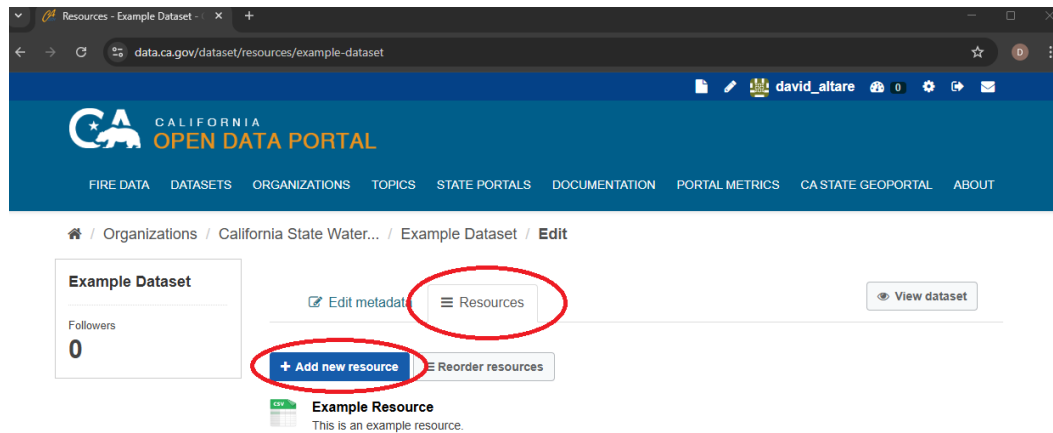
Example Resource
This is an example resource. [Explore](#)

water

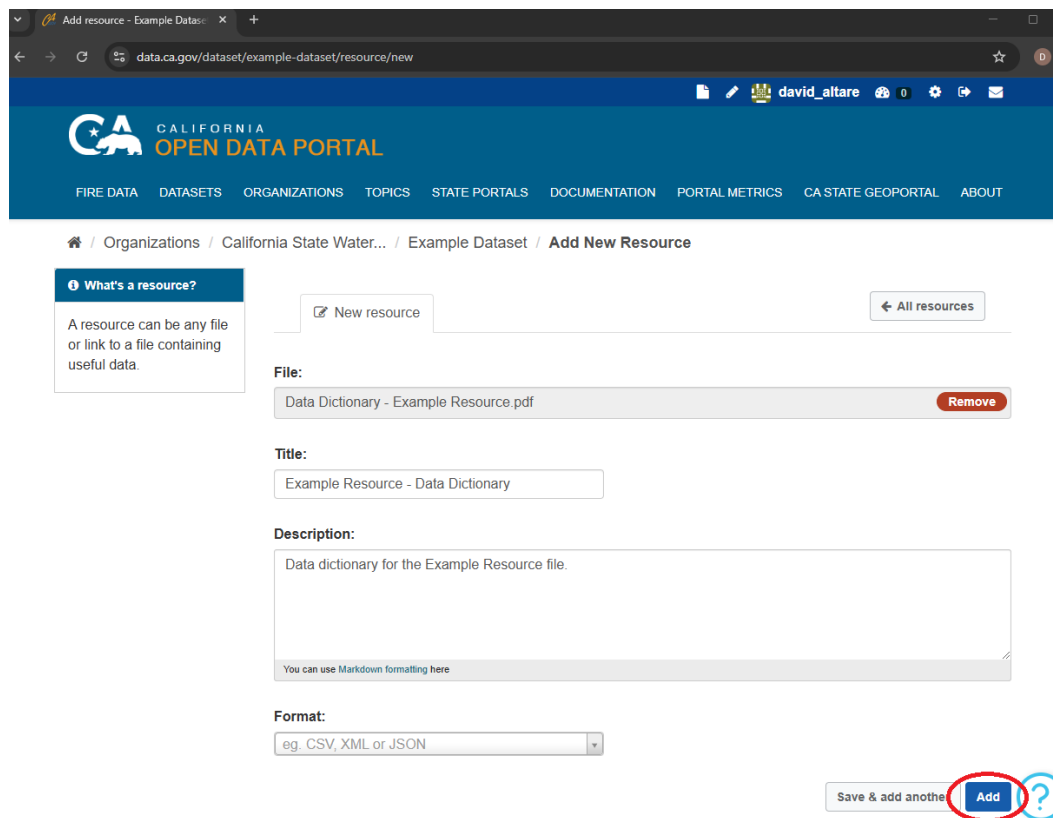
Additional Info [Download](#)

Field	Value
Public Access Level	Public
Rights	No restrictions on public use
Program Contact Name	Contact Name
Program Contact Email	waterdata@waterboards.ca.gov

Then click on the **Resources** tab, and click on the **Add new resource** button.



Upload the PDF file by clicking the **Upload** button, then click on the **Add** button.



- B. Add the data dictionary directly to the data resource by going to the data resource's page and clicking on the **Manage** button. (**NOTE:** After uploading a new data resource, before you can add data dictionary information you'll need to wait until the data table appears on the resource's page, as shown in the screenshot below; this could take a few minutes, depending on the size of the resource.)

The screenshot shows the 'Example Resource' page on the California Open Data Portal. The page has a blue header with the portal logo and navigation links. Below the header, the breadcrumb trail is 'Organizations / California State Water... / Example Dataset / Example Resource'. The main content area features a 'Manage' button circled in red, along with 'Setup for Reporting', 'DataStore', 'Download', and 'Data API' buttons. Below these buttons, the URL is displayed: <https://data.ca.gov/dataset/862443bb-e5c3-4ee0-9bac-44771c406b09/resource/194a9af0-4111-4472-a64c-b891b0652f2f/download/penguins...>. Below the URL, there is a 'Data Table' tab and a 'Canonical View' button. Further down, there are 'Add Filter', 'Show 10 entries', 'Hide/Unhide Columns', and a search bar. At the bottom, a table displays the first three entries of the dataset.

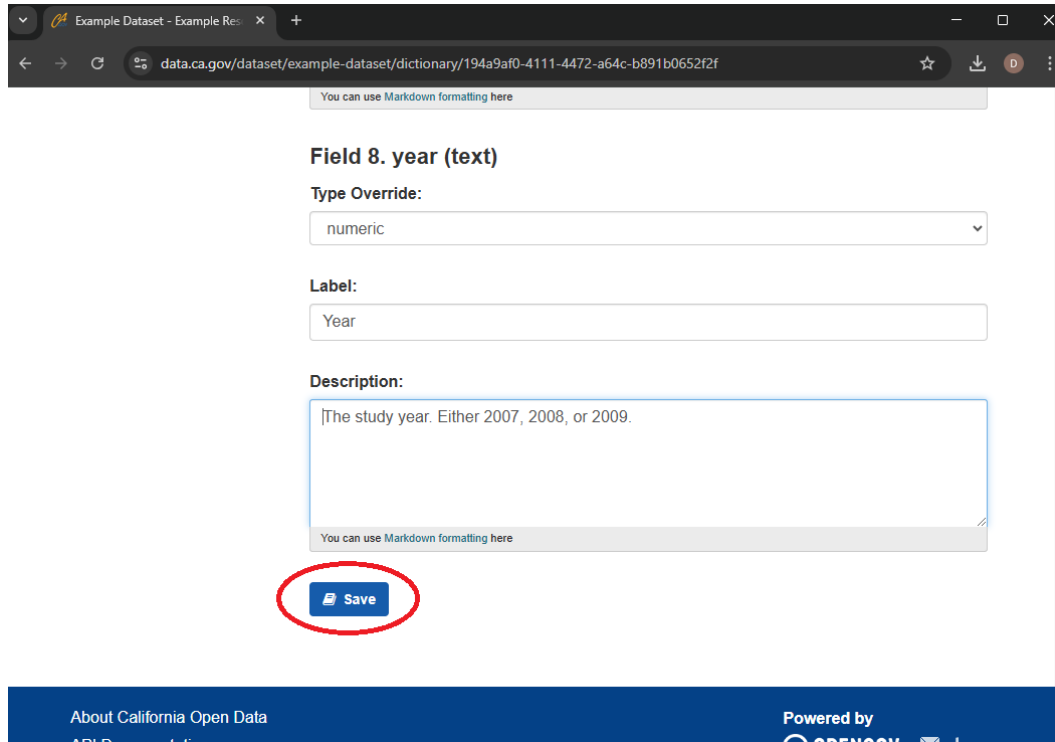
_id	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
1	Adelie	Torgersen	39.1	18.7	181	3750	male	2007
2	Adelie	Torgersen	39.5	17.4	186	3800	female	2007
3	Adelie	Torgersen	40.3	18	195	3250	female	2007

Go to the **Data Dictionary** tab, then fill in the available information for each field in the data resource.

The screenshot shows a web browser window with the URL `data.ca.gov/dataset/example-dataset/dictionary/194a9af0-4111-4472-a64c-b891b0652f2f`. The page header includes the California Open Data Portal logo and navigation links: FIRE DATA, DATASETS, ORGANIZATIONS, TOPICS, STATE PORTALS, DOCUMENTATION, PORTAL METRICS, CA STATE GEOPORTAL, and ABOUT. The breadcrumb trail is: Home / Organizations / California State Water... / Example Dataset / Example Resource / Edit. On the left, a sidebar shows 'Example Resource' with a 'Format' dropdown set to 'CSV'. The main content area has tabs: 'Edit resource', 'DataStore', 'Data Dictionary' (highlighted with a red circle), 'Setup for Reporting', and 'Views'. Above the 'Data Dictionary' tab are buttons for 'All resources' and 'View resource'. The 'Data Dictionary' tab contains the following fields:

- Field 1. species (text)**
- Type Override:** A dropdown menu.
- Label:** A text input field.
- Description:** A large text area with a note at the bottom: 'You can use Markdown formatting here'.

Click the **Save** button when finished.



The screenshot shows a web browser window with the URL `data.ca.gov/dataset/example-dataset/dictionary/194a9af0-4111-4472-a64c-b891b0652f2f`. The page displays a form for editing a dataset field. At the top, a notification bar says "You can use Markdown formatting here". The form is titled "Field 8. year (text)". It includes a "Type Override:" dropdown menu set to "numeric", a "Label:" text input field containing "Year", and a "Description:" text area containing "The study year. Either 2007, 2008, or 2009." Below the description field is another notification bar: "You can use Markdown formatting here". At the bottom of the form is a blue "Save" button with a floppy disk icon, which is circled in red. The footer of the page is a dark blue bar with the text "About California Open Data" and "Powered by" followed by the "OPENGOV" logo.

Field 8. year (text)

Type Override:

numeric

Label:

Year

Description:

The study year. Either 2007, 2008, or 2009.

You can use Markdown formatting here

 Save

About California Open Data
API Documentation

Powered by
OPENGOV

After saving the data dictionary information, the data dictionary will appear below the data table on the resource's page. (**NOTE:** you may need to update the resource's **DataStore** to make sure all of the changes to the data dictionary are applied to the resource; to do this, see the instructions in [Step 2](#) of the *Managing an Existing Dataset or Data Resource* section below.)

_id	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
1	Adelie	Torgersen	39.1	18.7	181	3750	male	2007
2	Adelie	Torgersen	39.5	17.4	186	3800	female	2007
3	Adelie	Torgersen	40.3	18	195	3250	female	2007
4	Adelie	Torgersen	NaN	NaN	NaN	NaN	NA	2007
5	Adelie	Torgersen	36.7	19.3	193	3450	female	2007
6	Adelie	Torgersen	39.3	20.6	190	3650	male	2007
7	Adelie	Torgersen	38.9	17.8	181	3625	female	2007
8	Adelie	Torgersen	39.2	19.6	195	4675	male	2007
9	Adelie	Torgersen	34.1	18.1	193	3475	NA	2007
10	Adelie	Torgersen	42	20.2	190	4250	NA	2007

Showing 1 to 10 of 344 entries

Previous 1 2 3 4 5 ... 35 Next

Column	Type	Label	Description
species	text	Species	The penguin's species. Either Adelie, Chinstrap, or Gentoo.
island	text	Island	The island in Palmer Archipelago, Antarctica where the penguin was observed. Either Biscoe, Dream, or Torgersen.
bill_length_mm	numeric	Bill Length (mm)	The penguin's bill length in millimeters.
bill_depth_mm	numeric	Bill Depth (mm)	The penguin's bill depth in millimeters.

NOTE: Setting the data type for each field by selecting a value in the **Type Override** box is especially useful if your users will be accessing the data via the API, but you'll need to ensure that all of the data in the given field is compatible with that data type before setting a data type other than 'text'. If there are problems with the data types, you can find more information by following the instructions in [Step 2](#) of the *Managing an Existing Dataset or Data Resource* section below, and going to the **DataStore** tab.

If you need to add data dictionary information for data resources with many fields and/or datasets with large numbers of resources, you can contact OIMA staff for assistance with automating the process of uploading that data dictionary information (to avoid the need for manual entry of the data dictionary information).

3 Managing an Existing Dataset or Data Resource

3.1 Update Dataset Metadata or Add Additional Data Resource

To add an additional data resource to an existing dataset, or to update the metadata for an existing dataset:

1. Go to the dataset and click on the **Manage** button.

The screenshot shows the California Open Data Portal interface. The top navigation bar includes links for FIRE DATA, DATASETS, ORGANIZATIONS, TOPICS, STATE PORTALS, DOCUMENTATION, PORTAL METRICS, CA STATE GEOPORTAL, and ABOUT. The user is logged in as 'david_altare'. The breadcrumb trail shows the path: Home / Organizations / California State Water... / Example Dataset. The main content area is titled 'Example Dataset' and includes a 'Test description' section. Below this, the 'Data and Resources' section lists two resources: 'Example Resource' (a CSV file) and 'Example Resource - Data Dictionary' (a PDF file). Each resource has an 'Explore' button. A search bar with the text 'water' is visible. The 'Additional Info' section includes a 'Download' button. On the left sidebar, the 'Example Dataset' section shows 0 followers and a 'Follow' button. The 'Organization' section displays the logo for the California State Water Resources Control Board and its website URL. The 'Manage' button in the top right corner of the dataset page is circled in red.

Example Dataset

Followers: 0

Follow

Organization: California State Water Resources Control Board

Website: <https://www.waterboards.ca.gov/> To preserve, enhance, and restore the state's water resources.

Dataset Topics Showcases Activity Stream Manage

Example Dataset

Test description

Data and Resources

Example Resource: This is an example resource. Explore

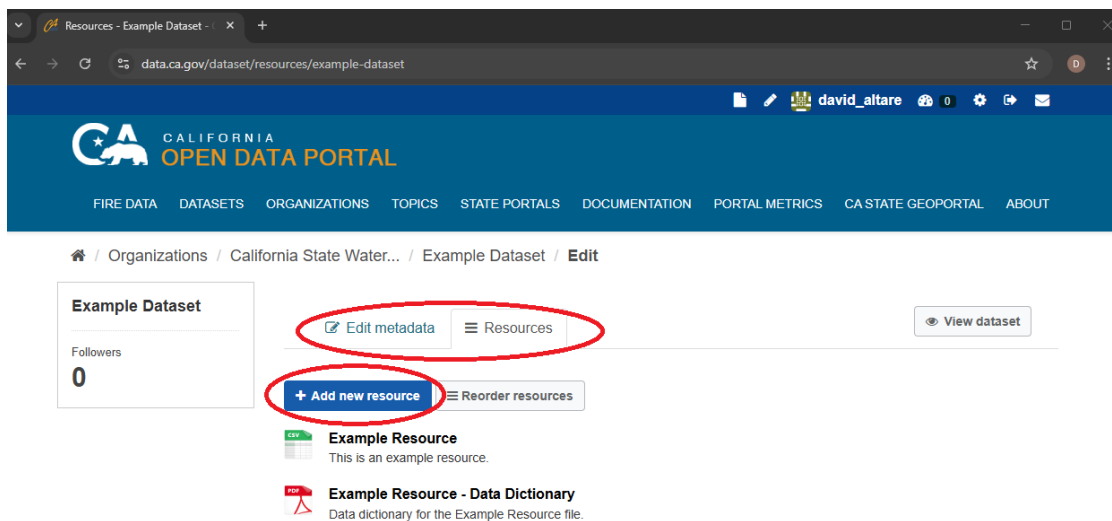
Example Resource - Data Dictionary: Data dictionary for the Example Resource file. Explore

water

Additional Info Download

Field	Value
-------	-------

2. To update the dataset's metadata, click on the **Edit metadata** tab. To add a new data resource to the dataset click on the **Resources** tab then click on the **Add new resource** button.



If you're adding a new data resource that doesn't already have an associated data dictionary available on the portal, you'll need to add a data dictionary for your resource. See [Step #6](#) above for information on how to do this.

3.2 Manage Existing Data Resource

To manage an existing data resource (i.e., an existing file within a dataset and/or the file's associated metadata):

1. Go to the data resource's page, then click on the **Manage** button.

The screenshot shows the California Open Data Portal interface. The breadcrumb trail is: Home / Organizations / California State Water... / Example Dataset / **Example Resource**. The page title is 'Example Resource'. Below the title, there are several buttons: 'Setup for Reporting', 'DataStore', 'Manage' (circled in red), 'Download', and 'Data API'. The URL is displayed as: <https://data.ca.gov/dataset/862443bb-e5c3-4ee0-9bac-44771c406b09/resource/194a9af0-4111-4472-a64c-b891b0652f2f/download/penguins...>. Below the URL, it says 'This is an example resource.' There are two tabs: 'Data Table' (selected) and 'Canonical View'. On the right, there are buttons for 'Fullscreen' and 'Embed'. Below these, there is an 'Add Filter' button. The 'Show' dropdown is set to '10' entries. Below that, it says 'Showing 1 to 10 of 344 entries'. On the right, there is a 'Search:' input field. The table has 9 columns: '_id', 'species', 'island', 'bill_length_mm', 'bill_depth_mm', 'flipper_length_mm', 'body_mass_g', 'sex', and 'year'. The first two rows of data are visible.

_id	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
1	Adelle	Torgersen	39.1	18.7	181	3750	male	2007
2	Adelle	Torgersen	39.5	17.4	186	3800	female	2007

2. Go to the **Edit Resource** tab to update the data file or update the resource's title or description, go to the **DataStore** tab to ensure that the data has been correctly uploaded to the database that feeds the resource's API, or go to the **Data Dictionary** tab to add information about each field and set the fields' data types.

data.ca.gov/dataset/example-dataset/resource/194a9af0-4111-4472-a64c-b891b0652f2f/edit

CA CALIFORNIA OPEN DATA PORTAL

FIRE DATA DATASETS ORGANIZATIONS TOPICS STATE PORTALS DOCUMENTATION PORTAL METRICS CA STATE GEOPORTAL ABOUT

Organizations / California State Water... / Example Dataset / Example Resource / Edit

Example Resource

Format CSV

All resources View resource

Edit resource DataStore Data Dictionary Setup for Reporting Views

File:

penguins.csv Remove

Title:

Example Resource

Description:

This is an example resource.

You can use Markdown formatting here

Format:

CSV

Delete Update Resource ?

4 Additional Resources

For more information about publishing open data resources on the California Open Data Portal, see the following resources:

- [California Open Data Publisher's Handbook](#)
- [California Open Data Portal Publisher's Guide](#)
- [California State Water Board's Open Data Publishing Guide](#)
- [California State Water Board's Recommendations and Guidelines for Data Dictionary Development](#)
- [California State Water Board's Open Data Handbook](#)